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Guest Editors
Special Issue VSI: AI4MSS
The Journal of Systems and Software

Dear Guest Editors,

We submit the manuscript “*Software-as-a-Graph: When Does Graph Learning Improve Cascade-Impact Ranking in Publish-Subscribe Systems?*” for consideration in the special issue.

This manuscript addresses the special issue topics “AI for Reliability and Dependability Analysis in Complex ICT Systems” and “Datasets, Benchmarks, and Evaluation Methodologies for AI Performance Models”. It evaluates when graph-learning techniques add value for pre-deployment cascading-failure analysis of publish-subscribe systems, compared with analytical rankings derived from the same dependency structure. It introduces a reference criterion to identify when an AI model’s agreement with a simulator reflects restatement of the simulator’s rule rather than predictive skill, a contribution to understanding the limitations of AI techniques, as the call highlights. The study also quantifies cost and estimated energy, showing that a training-free approximation computed in milliseconds ranks components nearly as a queue-flow simulator does (Spearman $\rho = 0.830$), a simulator that takes 12.7 CPU-hours to label the corpus.

The manuscript extends our RASSE 2025 paper with substantially more than 30% new technical content; the prior paper and a document detailing the improvements are included with the submission.

The manuscript is not under consideration elsewhere, and both authors have approved its submission. The datasets, harnesses and analysis scripts are archived on Zenodo (DOI 10.5281/zenodo.23045204).

Sincerely,

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Engineering, Istanbul Technical
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